Science practice booklet: Grade 3 expectations

Grade 3 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 3 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Living things

Standards practised: 3-LS1-1, 3-LS2-1, 3-LS3-1, 3-LS3-2, 3-LS4-1, 3-LS4-2, 3-LS4-3, 3-LS4-4

A worked example

Maryam sees a baby Arabian oryx standing next to its mother. Both have a pale coat and long, straight horns. Is the pale coat an inherited trait or a learned one? Explain.

1. An inherited trait is passed from parents to their young, like the colour of a coat.
2. The baby oryx did not have to learn to have a pale coat. It was born with it.
3. So the pale coat is an inherited trait.

Inherited: the baby is born with the coat its parents have.

Questions

1. Name the stages in the life cycle of a falcon, in order.

2. A butterfly starts as an egg. What stage comes after the caterpillar?

3. Put the life of a date palm in order: adult palm with fruit, seed, young palm, seedling.

4. Arabian oryx live in herds. Give one way that living in a group helps an oryx.

5. Which is an inherited trait: the colour of a child's eyes, or knowing how to ride a bicycle?

6. A falcon is trained to return to its owner's glove. Is this trait inherited or learned?

7. Two seeds from the same date palm are planted. One gets plenty of water and sunlight, the other gets very little. Which grows taller, and what does this show?

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8. Hamdan finds the fossil of a sea shell in a rock in the UAE desert, far from the sea. What might this tell us about that place long ago?

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9. A camel has wide, padded feet. How does this help it live in the desert?

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10. Mangrove trees grow in salty water on the coast. Their roots stick up out of the mud. Why is this useful?

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11. A small pond in a wadi dries up in a very hot summer. What might happen to the animals that lived there?

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12. Explain why the pale sandy coat of an Arabian oryx helps it survive in the desert.

Section 2: Earth and space

Standards practised: 3-ESS2-1, 3-ESS2-2, 3-ESS3-1

A worked example

The forecast for Tuesday in Dubai says 38 degrees and sunny, with no rain. Is that weather or climate? Explain.

1. Weather is what the air is like in one place on one day or over a short time.
2. Climate is the pattern of weather over many years.
3. A forecast for one day is about weather.

Weather: it describes one day.

Questions

1. What is the difference between weather and climate?

2. Which describes climate: "It rained this morning in Fujairah" or "Summers in the UAE are usually hot and dry"?

3. Which tool measures how much rain has fallen?

4. Which tool measures how hot the air is?

5. Rainfall in millimetres from Monday to Friday was: 0, 0, 5, 12, 0. On which day did it rain the most?

6. Using the same rainfall data, how much rain fell in the five days altogether?

7. The usual daytime temperature in one city is 22 degrees in January and 41 degrees in July. Which month is hotter, and which season is July in?

8. Describe the climate of most of the UAE in a few words.

9. A sandstorm is coming. Give two ways to stay safe.

10. After very heavy rain, water rushes down a dry wadi. Why should people stay away from it?

11. In July the weather can be dangerously hot. Give two ways to stay safe from the heat.

12. Each August for the last five years the top temperature in Abu Dhabi was above 40 degrees. What do you expect for next August? Explain.

Section 3: Matter, forces and energy

Standards practised: 3-PS2-1, 3-PS2-2, 3-PS2-3, 3-PS2-4

A worked example

Salem and Khalid pull a rope in opposite directions in a tug of war. They pull with the same strength. What happens to the rope?

1. Each boy pulls one way with a force.
2. The two forces are the same size and in opposite directions, so they are balanced.
3. Balanced forces do not change the motion of the rope, so it stays still.

The rope does not move.

Questions

1. What do we call two forces that are the same size and push or pull in opposite directions on an object?

2. In a tug of war, a team pulls much harder than the other team. Which way does the rope move?

3. A book rests on a table and does not move. Which two forces act on it?

4. Layla kicks a ball that is lying still. What does the kick do to the ball's motion?

5. Why does a ball that you drop fall to the ground?

6. A swing moves forward and back. One full swing takes 2 seconds. About how many full swings will it make in 10 seconds?

7. Omar notices that a toy car rolled down a ramp stops at about the same place each time he tries. How can he use this pattern?

8. Which of these will a magnet pick up: a steel paperclip, a plastic spoon, a wooden pencil?

9. Can a magnet pull a paperclip without touching it? Explain.

10. What happens when you bring the north ends of two magnets close together? What if you bring a north end and a south end together?

11. Hind wants a cupboard door to stay shut without a lock. How could she use two small magnets?

12. Two magnets keep pushing each other away when Noura tries to join them. What can she do to make them stick together? Explain.

Answer guide for parents

Section 1: Living things

1. Egg, chick, young falcon, adult falcon.
2. The chrysalis (pupa), and then the adult butterfly.
3. Seed, seedling, young palm, adult palm with fruit.
4. Any one of: more eyes to spot danger, protection for the young, or finding food and water together.
5. The colour of the eyes. Riding a bicycle is learned.
6. Learned: the falcon is taught it by its trainer.
7. The one with plenty of water and sunlight. The environment can change how a living thing grows.
8. It was once covered by the sea.
9. The wide feet spread its weight so it does not sink into soft sand.
10. The roots take in air, because the mud around them holds very little air.
11. Some move to find water in another place, and some may not survive. Changes in a habitat affect the animals living in it.
12. Pale colours reflect sunlight, which helps it stay cooler, and the coat matches the sand so it is harder to spot.

Section 2: Earth and space

1. Weather is what the air is like on a given day. Climate is the usual pattern of weather in a place over many years.
2. "Summers in the UAE are usually hot and dry" (it is a pattern over many years).
3. A rain gauge.
4. A thermometer.
5. Thursday (12 mm).
6. 17 mm (5 + 12).
7. July is hotter, and it is in summer.
8. Hot summers, mild winters, and very little rain.
9. Any two of: stay indoors, close the windows and doors, cover the nose and mouth if outside, protect the eyes.
10. The fast water can carry people and cars away, even if it looks shallow.
11. Any two of: drink plenty of water, stay in the shade or indoors at midday, wear light clothes and a hat.
12. Probably above 40 degrees again, because weather patterns from past years help us predict what is likely to happen.

Section 3: Matter, forces and energy

1. Balanced forces.
2. Towards the team that pulls harder, because the forces are unbalanced.
3. Gravity pulls it down and the table pushes it up. They are balanced.
4. It starts the ball moving. A push is a force that can change motion.
5. Gravity, a pull from the Earth, pulls it downwards.

6. 5 swings (10 divided by 2). The pattern repeats in the same time.
7. He can predict where the car will stop the next time it is rolled the same way.
8. The steel paperclip.
9. Yes. A magnet's force works across a small gap, so the paperclip can jump to it.
10. Two north ends push apart (repel). A north and a south end pull together (attract).
11. Fix one magnet on the door and one on the cupboard so that opposite ends face each other. They attract and hold the door closed.
12. Turn one magnet around. Then different ends face each other, and different ends attract.